



Mechanism underlying the uprooting of taproot-type shrub species in the loess area of northeastern Qinghai-Xizang Plateau, China

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Abstract: Characteristics of root pullout resistance determine the capacity to withstand uprooting and the slope protection ability of plants. However, mechanism underlying the uprooting of taproot-type shrub species in the loess area of northeastern Qinghai-Xizang Plateau, China remains unclear. In this study, a common taproot-type shrub, *Caragana korshinskii* Kom., in northeastern Qinghai-Xizang Plateau was selected as the research material. Mechanism of root-soil interaction of vertical root of *C. korshinskii* was investigated via a combination of a single-root pullout test and numerical simulation analysis. The results indicated that, when pulling vertically, axial force of the roots decreased with an increase in buried depth, whereas shear stress at root-soil interface initially increased and then decreased as burial depths increased. At the same buried depth, both axial force and shear stress of the roots increased with the increase in pullout force. Shear stress and plastic zone of the soil surrounding the root were symmetrically distributed along the root system. Plastic zone was located close to the surface and was caused primarily by tensile failure. In nonvertical pulling, symmetry of shear stress and plastic zone of the soil surrounding the root was disrupted. We observed larger shear stress and plastic zones on the side facing the direction of root deflection. Plastic zone included both shear and tensile failure. Axial force of the root system near the surface decreased as deflection angle of the pullout force increased. When different rainfall infiltration depths had the same vertical pulling force, root axial force decreased with the increase of rainfall infiltration depth and total root displacement increased. During rainfall infiltration, shear stress and plastic zone of the soil surrounding the root were prone to propagating deeper into the soil. These findings provide a foundation for further investigation of soil reinforcement and slope protection mechanisms of taproot-type shrub species in the loess area of northeastern Qinghai-Xizang Plateau and similar areas.

Keywords: Qinghai-Xizang Plateau; shrub roots; pullout test; numerical simulation; mechanism; root-soil interaction

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1 Introduction

Northeastern Qinghai-Xizang Plateau is located at the intersection between sub-humid temperate continental monsoon climate zone, the inland arid zone, and the high-cold area in China. It is an area with complex and sensitive climatic and environmental changes (Liu et al., 2021). Loess is widely distributed in this area and is characterized by a high sediment thickness, high salt content,

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vertically developed joints, loose structure, low shear strength, and strong water sensitivity (Fu et al., 2016; Zhuang et al., 2016). Due to the neotectonic movement, the loess in this area has been strongly eroded and cut, resulting in extensive development of loess slopes (Zhuang et al., 2016). Additionally, the slopes in this area are characterized by steep gradients and sparse vegetation, resulting in shallow rainfall infiltration. This phenomenon leads to frequent shallow landslides with irregular distributions, minor initial deformation, and extensive coverage (Liu et al., 2022). Landslides pose a significant threat, and predicting them is challenging, therefore, it is imperative to take appropriate engineering measures.

Previous studies have shown that plant roots play a crucial role in preventing and controlling soil erosion, shallow landslides, and other geological disasters. Plant root serves as an effective means of preventing and controlling shallow loess landslides (Liu et al., 2021). However, due to the shallow depth of these loess landslides, typically 2.0–3.0 m (Liu et al., 2022), the roots of herb species are primarily distributed within 1.0–1.5 m below the surface (van Beek, 2005; Vergani et al., 2017). Consequently, the key to achieving ecological protection for these shallow loess landslides is the root-anchoring effects of taproot-type woody plants with deep vertical roots. Common shrub species in this area include *Caragana korshinskii* Kom., *Tamarix chinensis* Lour., and *Hippophae rhamnoides* Linn (Liu et al., 2022). Notably, *C. korshinskii* and several other shrub species of the same genus, including *Caragana microphylla* Lam. and *Caragana opulens* Kom., possess taproot-type roots consisting of a deep taproot and numerous lateral roots. These shrub species have a strong adaptability, cold and drought tolerance, wind erosion resistance, and ease of reproduction. They are highly suitable for windbreak, sand fixation, soil and water conservation, and slope stabilization purposes (Hu et al., 2013; Liu et al., 2021).

Pullout resistance of plant roots is a crucial parameter for directly assessing the stability of the plant uprooting resistance and indirectly reflects windbreak, sand fixation, soil and water conservation, and slope stabilization capacities (Stokes et al., 1996; Stokes et al., 2005; Stubbs et al., 2019). The greater the pullout resistance of the root system, the smaller the peak displacement, and the more significant the stability against uprooting and the effect of soil consolidation and slope protection (Ji et al., 2018; Liu et al., 2022). Numerous studies have been conducted to investigate the characteristics of root pullout resistance and its influencing factors through *in situ* pullout tests, remolded soil pullout tests, and numerical simulation analyses. These studies aimed to further explore the mechanisms of uprooting. Previous research has shown that the pullout resistance of plant roots is primarily influenced by the type of the root system, root morphological indices, aboveground growth indices, tensile mechanical properties of individual roots, plant growth period (Burylo et al., 2009; Zhang et al., 2012; Yang et al., 2017; Yang et al., 2018; Liu et al., 2021), and physical and mechanical properties of the soil (such as the moisture content, dry density, shear strength, and soil type) (Moore, 2000; Dupuy et al., 2005; Mickovski et al., 2007; Burylo et al., 2009; Stokes et al., 2009; Schwarz et al., 2011; Zhang et al., 2012; Su et al., 2020; Zhang et al., 2020).

Mechanism of root-soil interface interaction is a pivotal aspect in studying the characteristics of the root pullout resistance. Root-soil interface exerts shear stress under pullout force, allowing plants to resist uprooting (Ennos, 1990). Previous studies have explored the mechanism of this interaction based on limit equilibrium theory and elastic-plastic mechanics theory. According to limit equilibrium theory, researchers conceptualized roots as a full-length bonded bolt to investigate their mechanical interaction with surrounding soil. Magnitude of the root anchorage force can be determined by accumulating shear forces at main root and each lateral root-soil interface (Liu et al., 2021). However, as root and soil are elastic-plastic materials, mechanical transfer and interaction between them is a nonlinear process when roots resist pullout forces (Ennos, 1989; Norris, 2005). Shear strength at root-soil interface gradually activates along the length of root (Mickovski et al., 2007). Therefore, limit equilibrium theory fails to provide a detailed explanation of the mechanism of root-soil interaction. Because this theory fails to determine load distribution characteristics and failure modes at root-soil interface during pullout resistance, previous studies have employed elastic-plastic mechanics theory to investigate the

mechanism of root-soil interface interaction (Ennos, 1989; Ennos, 1990). These studies have revealed characteristics of the strain distribution, axial force within root, and shear stress at root-soil interface during pullout resistance, providing a deeper understanding of the mechanisms involved in root-soil interaction.

Due to the intricate nature of root distribution and challenges in measuring stress and deformation in root-soil system, previous studies have focused on examining pullout resistance characteristics of plant roots using shallow root or model pullout tests (Mickovski et al., 2007). Additionally, theoretical and numerical simulation analyses have been utilized to further investigate the mechanisms of root-soil interaction (Ennos, 1989; Stokes, 1999; Dupuy et al., 2005; Dupuy et al., 2007). Few studies, however, have conducted single-root pullout experiments to examine the roots of woody plants that have straight roots during mature stage to obtain stress characteristics of these roots under different pullout forces. Additionally, studies are lacking, which combine these experimental results with numerical simulation analysis methods to obtain the stress characteristics of soil around roots and systematically discuss the mechanical mechanism of root-soil interaction under different pullout conditions. In summary, to improve our knowledge of uprooting mechanism for taproot-type shrub species in the loess area of northeastern Qinghai-Xizang Plateau, China, we selected the dominant shrub, *C. korshinskii*, in this area as research material, and conducted single-root pullout tests and numerical simulation analysis. The objectives of this study were: (1) to clarify the characteristics of stress of a vertical shrub root and soil surrounding the root during vertical pullout process; (2) to reveal the characteristics of stress of a vertical shrub root and soil surrounding the root under a non-vertical pullout force; and (3) to investigate the effect of rainfall infiltration on the characteristics of stress of a vertical shrub root and soil surrounding the root during vertical pullout process. The findings of this study provide a foundation for future studies on soil reinforcement and slope stabilization efforts using taproot-type shrub species in similar loess area.

2 Materials and methods

2.1 Study area

Study area is located on northeastern Qinghai-Xizang Plateau, China (36°44′02″N, 101°38′07″E; 2750–4100 m a.s.l.), which has a typical plateau continental climate (Long et al., 2018; Wang et al., 2022). Annual average temperature is 6°C, and average annual precipitation is approximately 350 mm, which mainly occurs between June and September (Liu et al., 2021). Most of precipitation in northeastern Qinghai-Xizang Plateau occurs in the form of short-term rainstorms and showers (Liu et al., 2021). In recent years, due to the influence of the Pacific subtropical anticyclone, there has been a notable increase in extreme climate events in this area, leading to a significant increase in precipitation (Wang et al., 2022). Consequently, this region has experienced frequent geological disasters, as well as increasingly severe ecological and engineering geological issues.

2.2 Single-root pullout test

Roots of *C. korshinskii* used in the test were collected in the study area in May 2023, which grown on a slope of village road. As the village road was being widened, roots of *C. korshinskii* were exposed, making them easy to collect. Previous studies by Liang et al. (2023) showed that average root diameters of middle-age and mature *C. korshinskii* roots were 17.40 (± 7.40) and 22.00 (± 4.80) mm, respectively, and root distribution depths reached 4.60 (± 0.80) and 5.50 (± 1.00) m, respectively. Based on this information, we selected a single root with an average root diameter greater than 20.00 mm as the research object under this root diameter condition, the strain gauge could be well attached to root surface to ensure normal functioning. Furthermore, because of limitations of test equipment and technical means, the maximum excavation limit of the test hole during pullout test was 1.50 m. Therefore, the maximum root depth was set at 1.50 m. Additionally, to study root stress characteristics of *C. korshinskii* root system under different

burial depths, we conducted single-root pullout tests under 0.80 and 1.20 m buried depths. Thus, three straight single-root segments with lengths of 1.10, 1.50, and 1.80 m and a minimum diameter of greater than 20.00 mm were selected as test samples for single-root pullout tests (Fig. 1a). Corresponding depths for these three single roots in pullout tests were 0.80, 1.20, and 1.50 m, respectively.

After collecting roots in the field, we sealed and wrapped them with plastic wrap and brought them back to laboratory to complete single pull test within 1–2 d. We did not conduct repeated experiments for each embedding depth for two main reasons: First, it was challenging to collect roots with similar diameters and lengths in the field. Second, after a single-root pullout test, it became difficult to reattach strain gauges to the root epidermis in an orderly and secure manner because of adhesion of strain gauges to the root surface and friction damage. Furthermore, friction strength of root-soil interface changed, which made it impossible to reuse the same single root for further testing.

Single-root pullout tests were conducted at a flat loess site located on western Qinghai University. Soil at the test site was silty. Physical properties of the soil are summarized in Table 1. Four resistive strain gauges were attached to each root at intervals of 0.20, 0.30, and 0.40 m (Fig. 1a).

Table 1 Soil physical properties in the study area

Depth (m)	Natural density (g/cm ³)	Moisture content (%)	Salt content (%)	Liquid limit (%)	Plastic limit (%)	Coefficient of uniformity
0.00–2.00	1.40±0.08	7.56±1.21	0.59±0.11	24.08±0.66	17.34±0.38	4.73±2.58

Note: Mean±SE.

Before single-root pullout tests, three vertical boreholes with a diameter of 0.30 m and depths of 0.80, 1.20, and 1.50 m were bored using a spiral drilling rig. Corresponding single-root segments with attached resistive strain gauges were then vertically installed in the center of each borehole. Area surrounding root segment in the borehole was filled with soil using a layered compaction method, achieving a density of 1.40 g/cm³. A self-constructed instrument was used for *in situ* pullout tests. Instrument was composed of a triangular bracket, a hand chain hoist, a clamp, a tension sensor (measurement tension of 30.00 kN and accuracy of 0.5%), a guyed displacement sensor (measurement length of 100.00 cm and accuracy of 0.5%), a data acquisition system, a laptop, and batteries. *In situ* pullout test instrument is shown in Figure 1b. To prevent deformation and movement of triangular bracket caused by reactive forces, we placed bricks at the bottom of triangular bracket to enhance its stability. For strain measurement of root segment, a static strain gauge was used. Pullout tests were conducted using step-by-step loading. Before each *in situ* pullout test, a clamp was attached to root segment just above ground surface. Hand chain hoist was then stretched tightly, ensuring that hand chain hoist remained on the same plumb line as plant stem, without applying tension to root segment. During the test, FLAC3D software was used to initialize tension sensor and displacement sensor and to activate the data acquisition system. Chain was then manually operated at a constant speed to apply different levels of pullout force to root segment. During the test, initial pullout force and pullout force increment were 1/10 of the estimate maximum pullout force of root segment. Estimated maximum pullout force ($F_{\max}$) exerted on root segment was obtained using Equation 1 (Liu et al., 2023).

$$F_{\max} = c_{rs} \pi D I + \tan \varphi_{rs} K_0 \pi D \gamma I^2 / 2, \quad (1)$$

where c_{rs} is the root-soil cohesion (kPa); D is the root diameter (m), and root diameters of the roots buried at depths of 0.80, 1.20, and 1.50 m are 23.00, 20.00, and 21.00 mm, respectively; I is the vertical burial depth of root (m); φ_{rs} is the comprehensive friction angle between root and soil (°); K_0 is the coefficient of the earth pressure at rest (0.64); and γ is the soil weight (kN/m³), which is 14.00 kN/m³ in this study.

After each pullout force application, root segment displacement was immediately recorded. Then, pullout force was held constantly, and displacement was continuously monitored for the

next 1, 6, and 10 min. If root segment displacement increment between 10 and 1 min under the same level of pullout force was less than 1.0 mm, strain on the root segment could be measured. Once measurement was completed, the next level of pullout force could be applied. If displacement increment exceeded 1.0 mm, the pullout force was held constantly, and further measurements were taken at 15, 30, and 60 min. If displacement between 60 and 6 min was less than 2.00 mm, root strain could be measured, and next level of pullout force could be applied. If displacement increment exceeded 2.00 mm, it was considered that ultimate load had been reached, and pullout test was terminated. The test was terminated if root displacement did not converge.

Failure of root-soil interface can be conceptualized as failure of microscopic soil layer encloses root (Zhang et al., 2002). Consequently, in this study, shear strength indices derived from direct shear tests on remolded soil were used as proxies for cohesion and friction angle of root-soil interface. For remolded soil shear test, we selected soil from the location of pullout test. Soil was taken back to the laboratory to dry. After passing through a 2.00-mm sieve, we prepared ring knife samples (3 groups, a total of 12) with a natural density of $1.40 (\pm 0.08) \text{ g/cm}^3$ and a natural water content of $7.56 (\pm 1.21)\%$. We used these samples to conduct an indoor direct shear test to obtain soil cohesion and internal friction angle. Detailed steps follow the geotechnical test procedures. These values were found to be $7.53 (\pm 2.30) \text{ kPa}$ and $26.71^\circ (\pm 1.84^\circ)$.

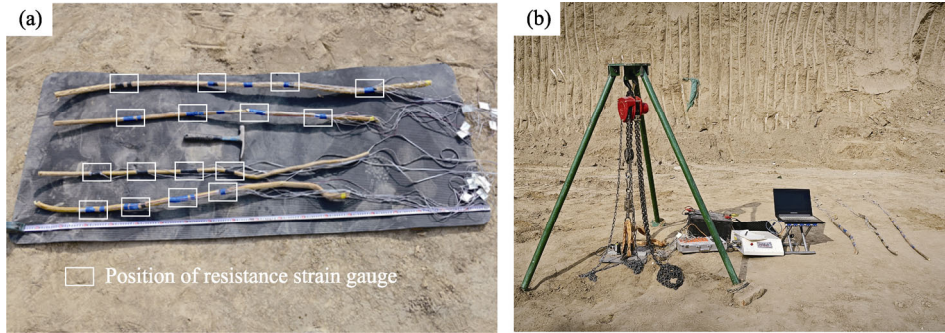


Fig. 1 Single-root pullout test. (a), single-root segments with resistive strain gauges; (b), self-constructed instrument used to conduct the pullout test. One of the roots in Figure 1a is the root system used in the preliminary experiment.

After pullout test, axial force at strain measurement point on root segment and average shear stress between two adjacent strain measurement points were obtained using Equations 2 and 3. Distribution patterns of axial force and shear stress at root-soil interface of vertical roots under varying pullout forces were calculated. However, accurately measuring strain data corresponding to root at the moment of pullout from soil is challenging, and thus, in this study, we focused solely on analyzing the distribution and variation patterns of axial force and shear stress at various positions along single root-soil interface under varying pullout forces before failure.

$$T_i = \varepsilon_i E S_i, \quad (2)$$

$$\tau_{i-i+1} = \frac{(\varepsilon_i S_i - \varepsilon_{i+1} S_{i+1}) E}{\pi D l}, \quad (3)$$

where T_i is the axial force at strain measurement point (kN); ε_i and ε_{i+1} are the axial strains at two adjacent strain measurement points; E is the elastic modulus of root (MPa); τ_{i-i+1} is the average shear stress at two adjacent strain measurement points (kPa); S_i and S_{i+1} are the cross-sectional areas of two adjacent strain measurement points (m^2); and D and l are the average diameter and root length of two adjacent strain measurement points, respectively (m).

2.3 Numerical simulation analysis

We used finite difference of FLAC3D software to conduct mechanical calculations. A three-dimensional model for simulating single-root pullout was developed. We verified the results

of pullout test and numerical simulation analysis, thereby enhancing validity of these research findings. Root, represented by a pile element with a 1.50-m burial depth and a 20.00-mm diameter, was positioned in the center of a semi-infinite ground space (length, width, and height of 3.00 m). Pullout force was exerted on the root model at the ground. Semi-infinite space, composed of solid elements and adhering to the Mohr-Coulomb elastic-plastic model, was meshed with cube units, which length, width, and height were 15.00 cm.

Numerical analysis encompassed three scenarios: vertical pullout, non-vertical pullout, and vertical pullout under rainfall infiltration. For vertical pullout, simulations were conducted with pullout forces of 0.42, 0.84, 1.26, and 1.80 kN. In non-vertical scenario, a force of 1.80 kN was applied to the root at deflection angles of 0°, 30°, and 60° relative to vertical direction. For vertical pullout under rainfall infiltration, root was subjected to a 1.80-kN force with infiltration depths of 0.00, 0.50, and 1.50 m. In this study, we set soil water content in different rainfall infiltration depths to saturated water content of the soil. Numerical analysis provided insights into the axial force distribution of vertical root and shear stress and plastic zone characteristics of surrounding soil under various conditions.

Physical and mechanical parameters of *C. korshinskii* root required for numerical simulation analysis, as reported by Shi et al. (2023), included elastic modulus, Poisson's ratio, and tensile strength, which were determined to be 297.10, 0.30, and 32.00 MPa, respectively. For numerical simulation analysis, cohesion and internal friction angle of root-soil interface were determined to be 7.53 (± 2.30) kPa and 26.71° ($\pm 1.84^\circ$), respectively. Referring to Han (2020), he found that cohesion and internal friction angle of saturated loess decreased by 47.50% and 7.70%, respectively, compared with their natural state values. Based on these findings, the adjusted cohesion and internal friction angle for root-soil interface of saturated loess were 3.95 kPa and 24.67°, respectively. Elastic modulus and Poisson's ratio of loess reported by Jiang (2022) were determined to be 22.00 MPa and 0.34, respectively, under natural conditions and 18.00 MPa and 0.30, respectively, under saturated conditions. Natural density and saturated density of loess were determined to be 1.40 and 1.77 g/cm³, respectively.

2.4 Statistical analysis

Through calculation and analysis of data obtained from single-root pullout test, we identified the relationship of axial force and shear stress with pullout load under different root lengths. We performed all statistical analyses using Origin v.8.0 software.

3 Results

3.1 Characteristics of root stress during *in situ* pulling

Under varying pullout forces and burial depths, the maximum axial force for a single root observed at the surface position is shown in Figure 2. As burial depths increased, the axial force of a single root gradually decreased. With increasing pullout force, axial force at different positions along a single root also increased, and upper part experienced a greater increase than the lower part. For roots buried at depths of 0.80 and 1.20 m, axial force was measured at all of the measurement points within root length range when initial force was low under pullout force. However, for single root buried at a depth of 1.50 m, axial force value was not measured at the lowest measurement point of 1.20 m under any level of pullout force. Results indicated that burial depths of root affected the mechanism of root-soil interaction during pullout.

Due to the zero shear stress on free surface above soil surrounding surface roots, shear stress at the root-soil interface of this position was zero, which adhered to the principle of shear stress equality. As shown in Figure 2, shear stress at root-soil interface initially increased and then decreased with increasing burial depths. The maximum shear stress occurred within the depth range of 0.10–0.30 m below ground surface under varying pullout forces. Notably, position of the maximum shear stress remained consistent for single root-soil interfaces at different burial depths, regardless of pullout force applied.

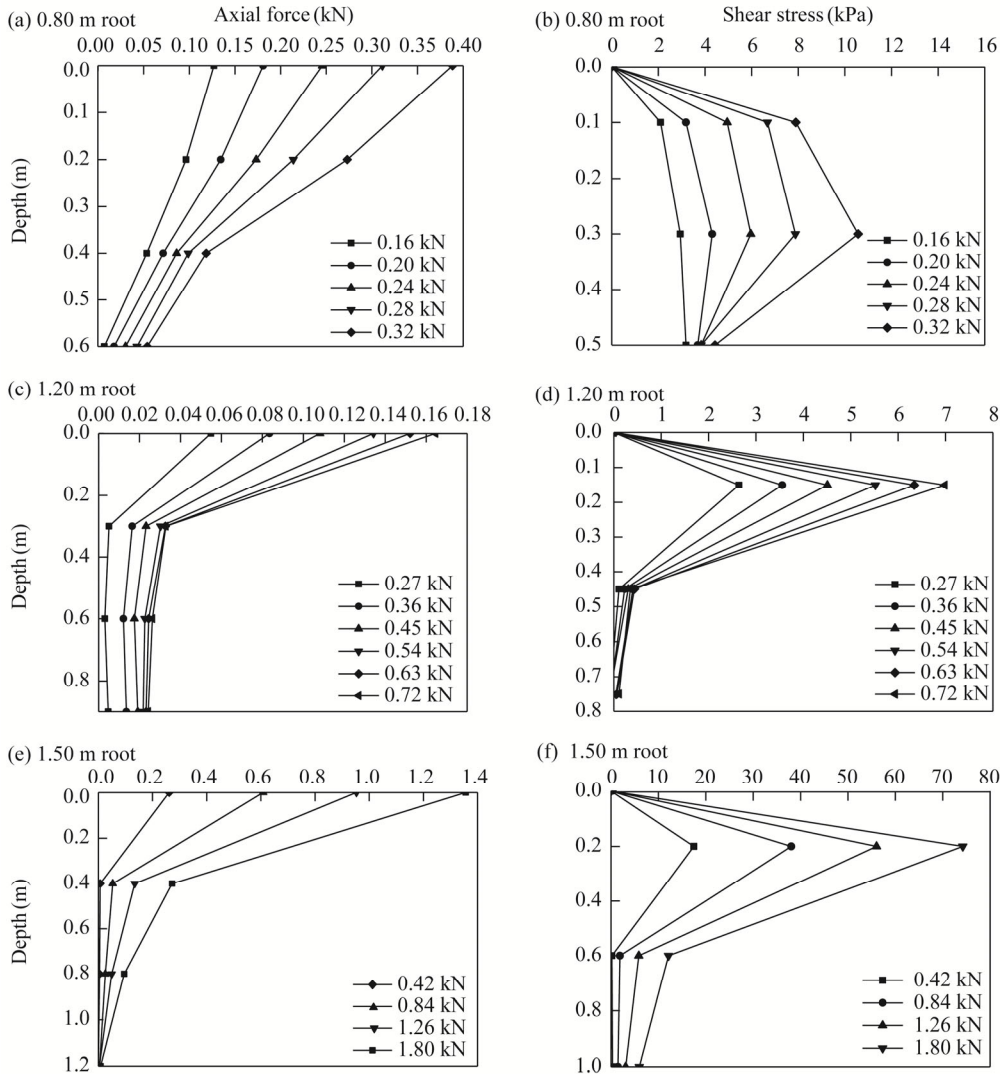


Fig. 2 Characteristics of axial force and shear stress of roots during pullout under different loads. (a), axial force distribution curve for 0.80 m root; (b), shear stress distribution curve for 0.80 m root; (c), axial force distribution curve for 1.20 m root; (d), shear stress distribution curve for 1.20 m root; (e), axial force distribution curve for 1.50 m root; (f), shear stress distribution curve for 1.50 m root.

3.2 Numerical simulation of mechanical characteristics of root-soil interaction

3.2.1 Stress characteristics of root and soil during vertical pullout force

Root axial forces with a buried depth of 1.50 m under different vertical pullout forces are shown in Figure 3. Results of numerical simulation analysis were similar to the change trend of single axial force for buried depth from *in situ* pullout test. As shown in Figure 4, under different pullout forces, shear stress and plastic zone of soil surrounding root exhibited symmetric distributions on both sides of root. As pullout load increased, shear stress and plastic zone surrounding root extended further toward both sides of root and deeper. Notably, Figure 4a–d showed that the maximum shear stress at root-soil interface gradually increased to 2.70, 5.70, 8.20, and 10.33 kPa when the pullout force was 0.42, 0.84, 1.26, and 1.80 kN, respectively. The maximum shear stress values occurred in the soil near surface, which was consistent with the distribution characteristics observed in pullout test. Furthermore, Figure 4e–h indicated that when root system was subjected to a vertical pullout force without reaching limit state, plastic zone of soil surrounding the root was primarily located around the upper part of root and close to surface position. Tensile failure

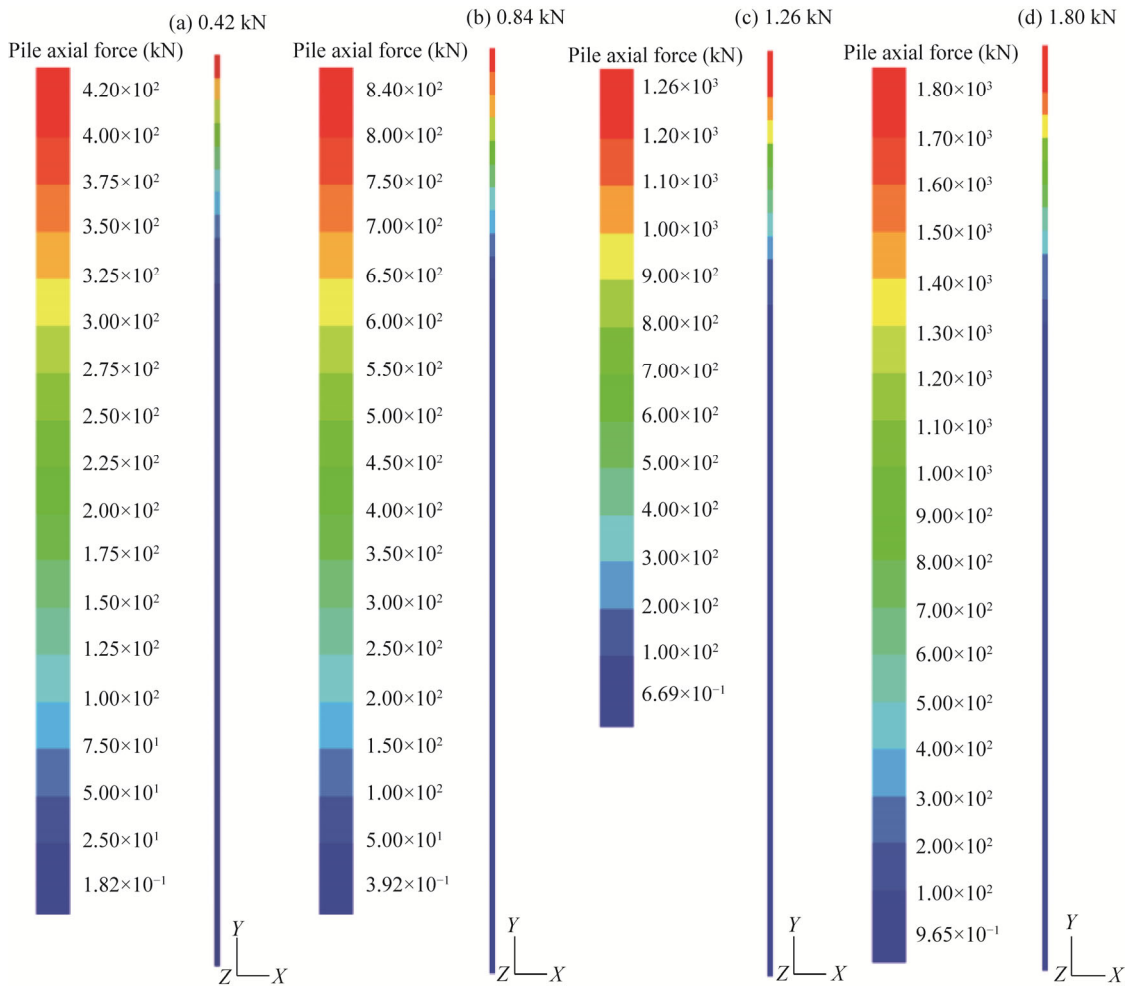


Fig. 3 Root axial force diagram under different vertical pullout forces. (a), 0.42 kN; (b), 0.84 kN; (c), 1.26 kN; (d), 1.80 kN. X, Y, and Z are the directions of length, width, and height of the model, respectively.

in this area exhibited an inverted conical distribution, and range of tensile failure area gradually increased as the pullout force increased.

3.2.2 Stress characteristics of root and soil surrounding root under a non-vertical pullout force

As shown in Figure 5, when subjected to non-vertical pullout forces with the same magnitude but different directions, variation characteristics of axial force of root still exhibited a gradual decrease with increasing root burial depths. Under non-vertical pullout force conditions, root axial force at surface position exhibited a decreasing trend compared with that observed under a vertical pullout force as deflection angle of pullout force increased. Specifically, when deflection angles were 30° and 60° , single-root axial forces at surface position were 1.64 and 0.90 kN, respectively, decreasing 8.89% and 50.00% compared with those observed under a vertical pullout force. Within the depth range of 0.00–0.60 m, axial force at a given position along root was the greatest when deflection angle was 30° , followed by 0° and 60° . However, below the depth of 0.60 m, root axial force tended to be consistent across different deflection angles.

As shown in Figure 6a and b, shear stress distribution characteristics of soil surrounding root under different pullout angles differed significantly from those observed under vertical pullout direction. When deflection angles were 30° and 60° , shear stress of soil surrounding root near surface on the side of pullout force gradually increased compared with that observed under vertical pullout force conditions and was higher than that on the side for the same burial

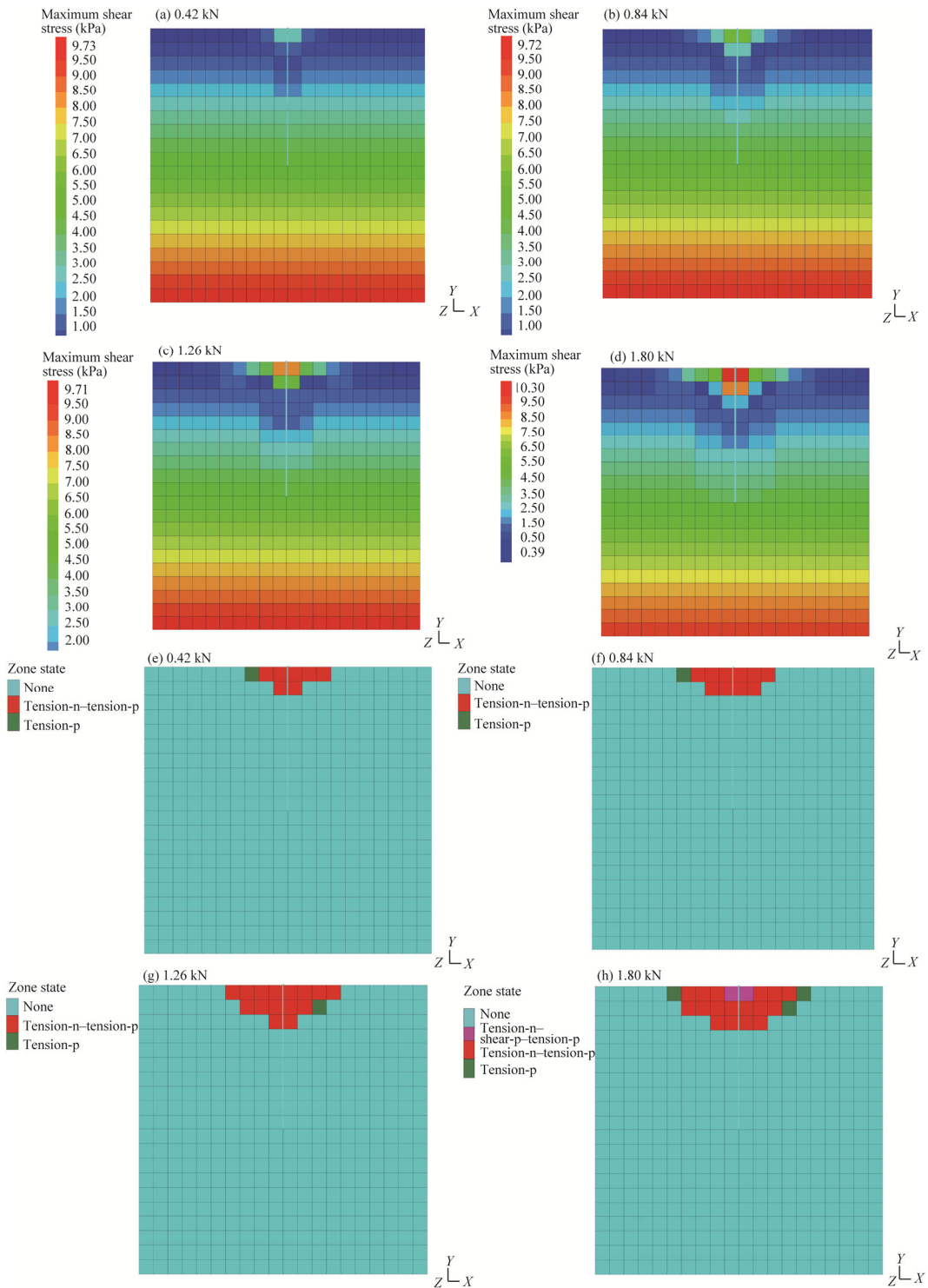


Fig. 4 Distribution characteristics of shear stress and plastic zone of soil surrounding root under different vertical pullout forces. (a)–(d), distribution characteristics of shear stress of soil surrounding root under different vertical pullout forces of 0.42, 0.84, 1.26, and 1.80 kN, respectively; (e)–(h), distribution characteristics of plastic zone of soil surrounding root under different vertical pullout forces of 0.42, 0.84, 1.26, and 1.80 kN, respectively. Tension-n, shear-p, and tension-p are the unit that is in tensile yield now, the unit that is in shear yield in the past, and the unit that is in tensile yield in the past, respectively.

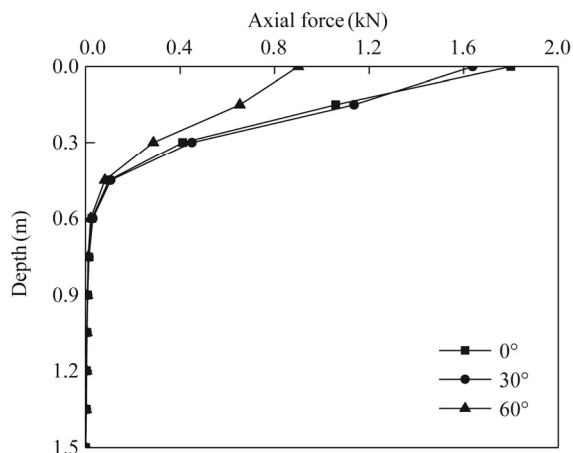


Fig. 5 Distribution of root axial force under different pullout angles

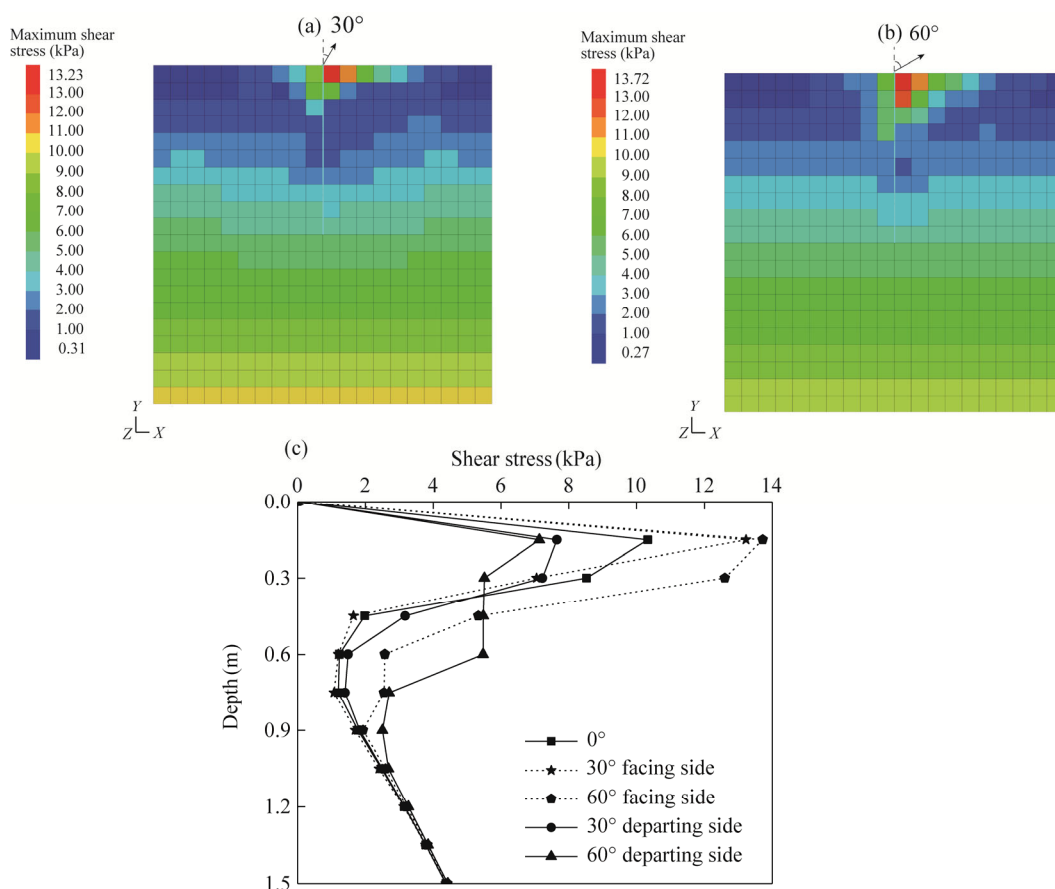


Fig. 6 Distribution of shear stress of soil surrounding root under different pullout angles. (a), 30°; (b), 60°; (c), shear stress of root-soil interface under different pullout angles and sides.

depth that deviated from pullout force. In Figure 6c, taking the burial depth of 0.15 m as an example, when deflection angles were 30° and 60°, shear stresses of root-soil interface nearer the side of pullout force were 13.23 and 13.72 kPa, respectively, increasing 28.07% and 32.82% compared with those observed under vertical pullout force conditions and increasing 72.94% and 92.43% compared with those deviating from pullout force, respectively. In the depth range of 0.45–1.05 m, root-soil interface shear stress on the side away from pullout force was higher than

that on the side near pullout force at the same position under the same deflection angle pullout force. Moreover, in this depth range, shear stress at root-soil interface on the side that deviated from pullout force at the same root position increased gradually as deflection angle of pullout force increased (Fig. 6c).

As shown in Figure 7, under non-vertical pullout conditions, plastic zone of the soil surrounding root exhibited an asymmetrical distribution. Plastic zone was noticeably shifted toward pullout force side. Close to surface, plastic zone on the side facing pullout force was larger than its counterpart on the opposite side. Furthermore, when a single root experienced a non-vertical pullout force, soil surrounding root near surface exhibited both tensile failure and shear failure (Fig. 7). This result differed significantly from failure characteristics observed in the soil under vertical pullout force conditions acting on a single root.

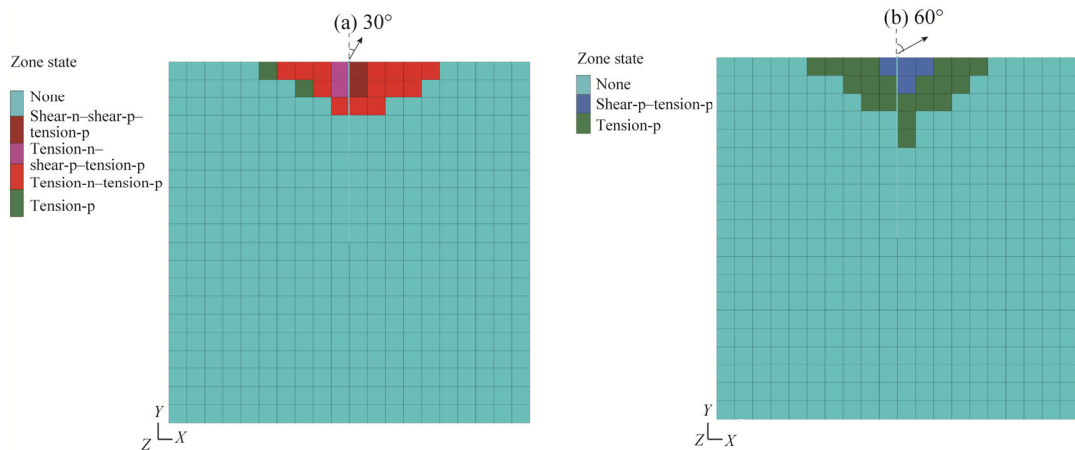


Fig. 7 Plastic zone of soil surrounding root under different pullout angles. (a), 30°; (b), 60°. Shear-n is the unit that is in shear yield now.

3.2.3 Stress characteristics of root and soil surrounding root under vertical pullout with rainfall infiltration depths

As shown in Figure 8, depth of rainfall infiltration played a significant role in determining distribution characteristics of root axial force under vertical pullout conditions. In the depth range of 0.00–0.80 m, except at surface position, at the same position, root axial force under the condition of a rainfall infiltration depth of 0.00 m was significantly higher than those under infiltration depths of 0.50 and 1.50 m. However, below a depth of 0.80 m, there was no significant distinction in axial force at the same position under different rainfall infiltration depths.

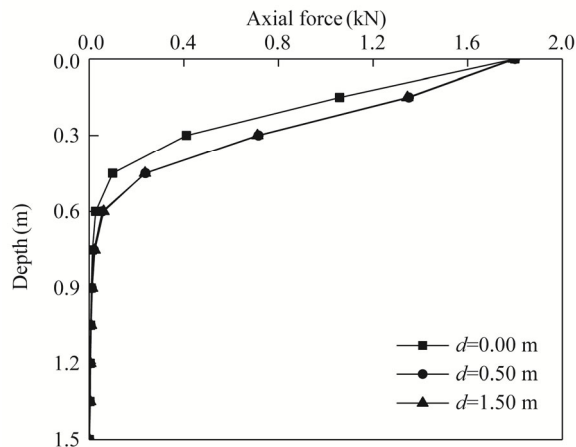


Fig. 8 Distribution of root axial force under different rainfall infiltration depths. d is the rainfall infiltration depth.

Under varying rainfall infiltration depths and the same vertical pullout force, shear stress distribution characteristics of soil surrounding root remained generally consistent, but there were notable differences in shear stress values at different soil locations (Fig. 9). Taking root-soil interface shear stress as an example (Fig. 9c), within the depth range of 0.00–0.30 m, root-soil interface shear stresses corresponding to infiltration depths of 0.50 and 1.50 m were relatively similar and were lower than that at the same position when infiltration depth was 0.00 m. However, below a depth of 0.50 m, shear stresses at root-soil interface for infiltration depths of 0.50 and 1.50 m were greater than that at the same position for an infiltration depth of 0.00 m. Below a depth of 0.60 m, shear stress of root-soil interface under condition of an infiltration depth of 1.50 m was significantly higher than that of root-soil interface at the same position when infiltration depth was 0.50 m.

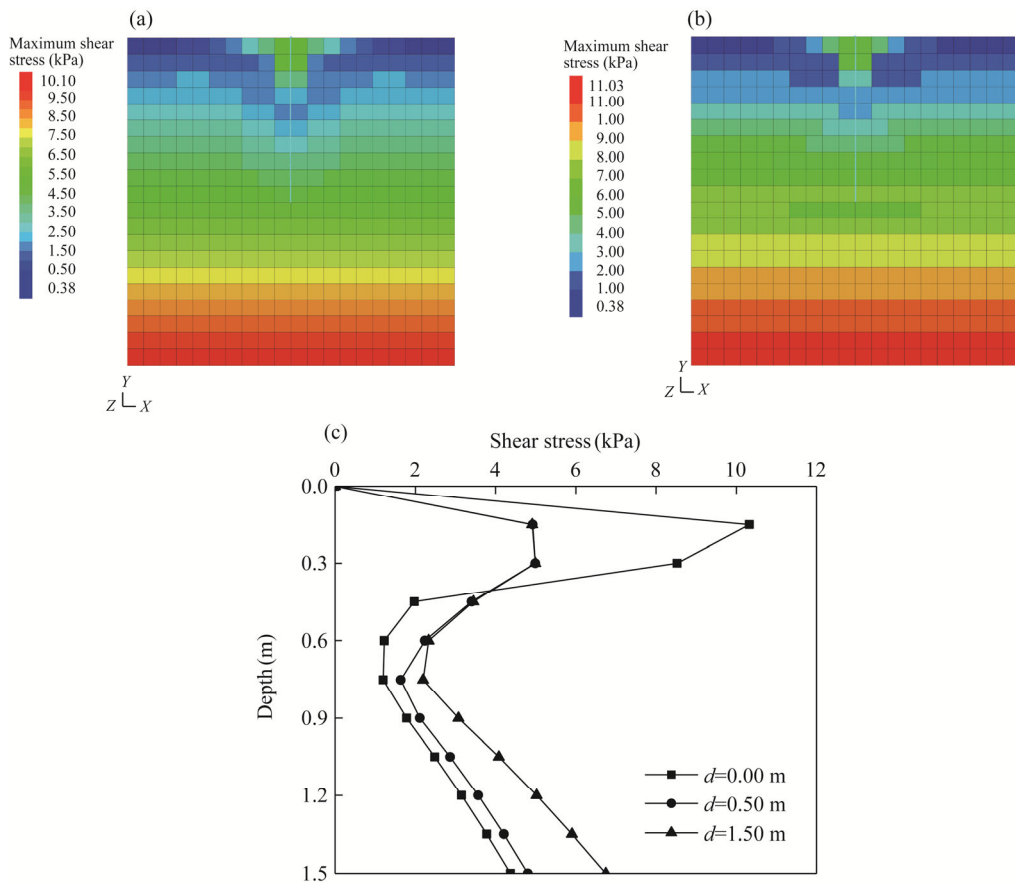


Fig. 9 Distribution of shear stress of soil surrounding root under vertical pullout and different rainfall infiltration depths. (a), 0.50 m; (b), 1.50 m; (c), shear stress of root-soil interface under vertical pullout and different rainfall infiltration depths.

As can be seen from Figure 10a and b, under identical vertical pullout forces, distribution of plastic zone in soil surrounding root remained consistent with that observed in the absence of infiltration. This zone was symmetrically distributed around single root and exhibited both shear failure and tensile failure. Additionally, Figure 10c and d demonstrated that for varying rainfall infiltration depths, the maximum root displacement occurred at soil surface. For infiltration depths of 0.50 and 1.50 m, the maximum root displacement increased 20.95% and 21.01%, respectively, compared with that of no infiltration.

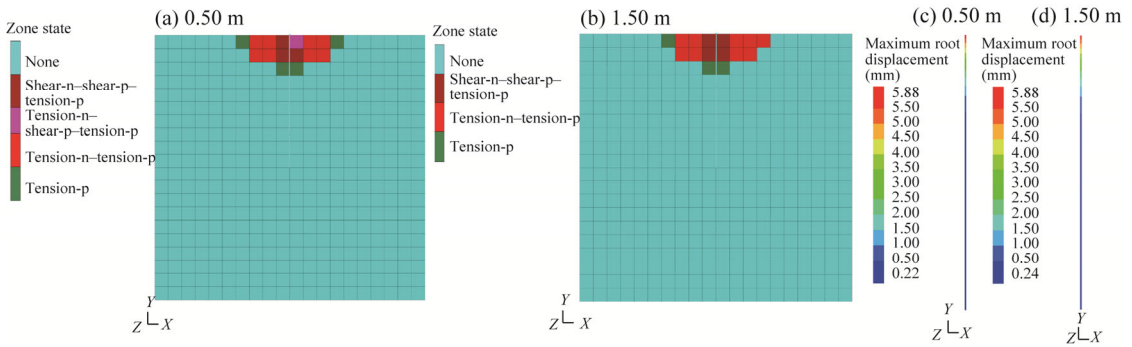


Fig. 10 Plastic zone of soil surrounding root under vertical pullout with rainfall infiltration depths of 0.50 m (a) and 1.50 m (b). Root displacement under vertical pullout for rainfall infiltration depths of 0.50 m (c) and 1.50 m (d).

4 Discussion

4.1 Mechanism of root-soil interaction under vertical pullout conditions

In this study, we integrated pullout tests with numerical simulation analysis to gain a deeper understanding of mechanical behavior that occurred between shrub roots and soil under vertical pullout conditions. The results of this study revealed that as pullout force gradually increased, both axial force of roots at different positions and shear stress at root-soil interface increased. Furthermore, plastic zone around roots continued to expand, extending to both sides and deeper below surface. This observation suggested that as pullout force intensified, axial force of root and shear stress at root-soil interface were progressively transferred downward. Additionally, shear strength at root-soil interface was gradually mobilized. You and Chen (2018) conducted a study on full-length bonded bolts and observed similar force distribution patterns.

Based on the above results and numerical simulation results, we summarized the mechanism for root-soil interaction under vertical pullout conditions. During initial phase of pullout (Fig. 4a), elastic deformation occurred primarily at the stressed end of root (Liu et al., 2022), leading to shear stress formation at root-soil interface in this vicinity (Ennos, 1989; Schwarz et al., 2010). As the load increased within elastic limit range, the maximum shear stress at root-soil interface near the stressed end also increased. As load continued to increase (Fig. 4b and c), when the maximum shear stress at root-soil interface at stressed end surpassed interfacial shear strength or shear strength of soil surrounding root, the interface or the soil in this area would enter the plastic deformation phase, leading to the local plastic slip deformation (Schwarz et al., 2010; Liu et al., 2022). Simultaneously, the maximum shear stress at root-soil interface would gradually extend deeper along the root system until all of the interfaces within root length range transferred into a plastic state (Ennos, 1989; Zhou et al., 1998). At this time, interfacial shear strength or shear strength of soil surrounding root will be reduced to their residual values, ultimately resulting in the root being pulled out of soil (Schwarz et al., 2011; Su et al., 2020). Additionally, because tensile strength of loess was significantly lower than its shear strength, shear stress at root-soil interface might cause tensile failure in soil surrounding root near surface, particularly under conditions of low self-weight stress.

Notably, in numerical simulation analysis, shear stress of soil surrounding root was a result of combined influence of relative root-soil dislocation and lateral soil pressure. As lateral soil pressure was directly proportional to burial depths of soil (Figs. 4a–d, 6, and 9), shear stress of soil surrounding root experienced an increase corresponding to deeper burial depths beneath mutual dislocation zone between root and soil. However, it should be noted that in pullout tests, root-soil interface shear stress was primarily related to axial root deformation. Therefore, measurement in the undeformed root area might not be accurate. As a result, shear stress did not increase with increasing burial depths at lower part of root, as shown in Figure 2b, d, and f. For

numerical simulation analysis, lateral soil pressure remained constant at each point along the root system under varying pullout tests. Therefore, differences in shear stress distribution characteristics were primarily due to root deformation. In conclusion, both *in situ* pullout tests and numerical simulation analysis effectively captured shear stress distribution characteristics at root-soil interface under pullout force conditions.

4.2 Influence of non-vertical pullout force on mechanism of uprooting

When plants are exposed to surface loads such as water, wind, and landslide forces, their roots below surface or beneath sliding area are subject to tensile stresses at a given angle (Liu et al., 2022). While previous research has primarily focused on differences in the maximum pullout resistance among roots at varying pullout angles (Stokes et al., 2007; Kamimura et al., 2012; Liu et al., 2013), there is a lack of comprehensive studies on the influence of non-verticality pullout force on the mechanism of uprooting.

The results of this study demonstrated that mechanism of single-root pullout resistance differed significantly when pullout angle varied from vertical. These differences were primarily manifested by factors such as axial force applied to root, shear stress within soil surrounding root, and extent of plastic zone. The reason for the differences could be contributed to the connection between root flexure deformation and changes in soil stress conditions surrounding roots when subjected to pullout forces at varying angles (Commandeur and Pyles, 1991; Su et al., 2020). Under a specific deflection angle, the root system might undergo flexural deformation toward pullout force side (Fig. 11). This deformation resulted in additional compressive stress being applied to soil surrounding root, leading to an increase in shear stress at root-soil interface. Conversely, shear stress at root-soil interface on the side farther away from pullout force decreased. Moreover, all of these values were higher than those at the same burial depth but on the side farther away from pullout force's direction. As deflection angle increased, shear stress at root-soil interface on pullout force's side increased, while shear stress on the opposite side gradually decreased. Within this depth range, soil surrounding root on the pullout force's side mainly experienced shear failure and tensile failure. Conversely, soil surrounding root on the side farther away from pullout force mainly underwent tensile failure.

Due to stiffness of the root system (Ennos, 1990), below a certain depth at surface, the root system might experience flexure deformation in an opposite direction of pullout force, as shown in Figure 11. Consequently, normal stress and shear stress on root-soil interface nearer pullout force in reverse deflection deformation range might increase, while those departing from deflection deformation might decrease. This phenomenon became more pronounced as deflection angle of surface pullout force increased. Therefore, within the depth range of 0.45–1.05 m, shear stress of root-soil interface on side farther away from pullout force was higher than that on the side nearer pullout force at the same position under the same deflection angle pullout force (Fig. 6c). As burial depths increased further, bending deformation of the root system decreased (Fig. 11). Therefore, in the depth range below 1.05 m (Fig. 6c), distribution characteristics of shear stress at root-soil interface under non-vertical and vertical pullout forces tended to be consistent. Under non-vertical pullout conditions, mechanical energy generated by pull force led to both elastic deformations of root and shear action at root-soil interface; additionally, it could induce bending and deformation of root (Commandeur and Pyles, 1991). Consequently, measured root axial force at pulling end was significantly smaller under non-vertical pulling force conditions than under vertical pulling conditions.

In summary, deflection pullout effect would change stress characteristics of soil surrounding root, and thus, soil surrounding root toward surface on the side nearer pullout force was more likely to be damaged. Simultaneously, pullout resistance provided by the root system was reduced. Therefore, when selecting woody plants for soil and water conservation and shallow landslide prevention, low shrubs should be selected as much as possible to reduce lateral force generated by surface loads such as water and wind power and to ensure that the plants had sufficient ability to maintain stability of withstand uprooting. In addition, some engineering measures can be adopted

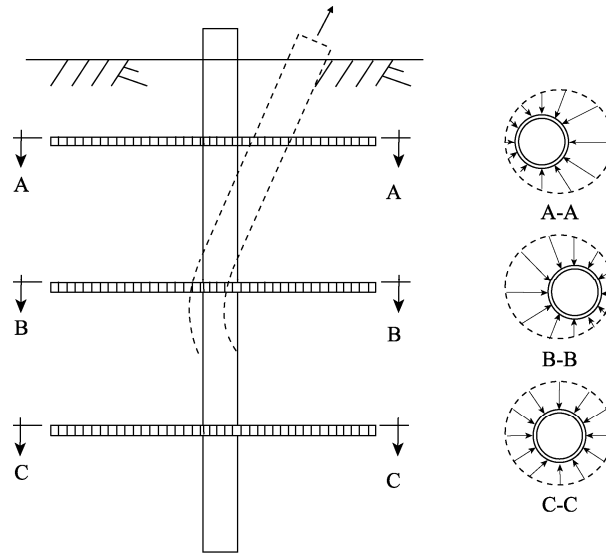


Fig. 11 Schematic diagram of root system stress under non vertical pull-out force. A-A, B-B, and C-C are the cross-sections of root system near pullout force, transitional root system, and root system far away from pullout force, respectively.

to improve the strength of soil surrounding roots within a certain range of ground surface to improve stability of withstanding uprooting.

4.3 Effect of rainfall infiltration on mechanism of vertical uprooting

In this study, we investigated interaction mechanism of root-soil interface under different infiltration depths. The results showed that under different rainfall infiltration depths and the same vertical pulling force, axial force of root system decreased with an increase in rainfall infiltration depth, while total root displacement increased. When rainfall infiltrated, shear stress and plastic zone of soil surrounding roots were more likely to propagate. This result suggested that increasing soil moisture content decreased shear strength at root-soil interface (Fan and Su, 2008; Han and Vanapalli, 2017), thus increasing relative stiffness of root system (Mickovski et al., 2007) and reducing the effect of soil consolidation and slope protection provided by plant roots (Mickovski et al., 2007; Zhang et al., 2020). Under the same pullout force, shear strength of root-soil interface in rainfall infiltration range was transferred more uniformly and rapidly. Root-soil interface entered plastic state more quickly, which allowed pullout load to be transferred to deeper positions in root-soil interface more rapidly, enabling deep interface to withstand greater shear stress. Differences in distribution characteristics of axial force and shear stress at root-soil interface under different infiltration depths were found (Figs. 8 and 9).

Root pullout resistance mechanism under rainfall infiltration had a direct impact on site stability of plants and anchoring effect of plant roots. Relevant studies have shown that in loess area of northeastern Xizang Plateau, root distribution depths of *C. korshinskii*, a principal shrub species used for soil conservation and slope stabilization, were approximately 3.61 (± 0.49), 4.60 (± 0.83), and 5.50 (± 1.00) m in the juvenile, middle-age, and mature stages, respectively (Liang et al., 2023). The maximum annual precipitation infiltration depth in loess area of the Qinghai-Xizang Plateau was about 1.00–3.00 m, and depth of shallow landslides often ranged from 2.00 to 3.00 m (Liu et al., 2023). Therefore, in this area, the effectiveness of root system in stabilizing slopes has depended on the depth of root distribution (growth period) and the depth of rainfall infiltration (potential slide depth). Only when root system was sufficiently long to extend below rainfall infiltration depth (potential sliding depth) and had a certain effective length could the root system effectively enhance the stability of landslide (Stokes et al., 2009; Liu et al., 2022).

In theory, length of roots penetrating rainfall infiltration depth (potential slip surface depth) should not be excessively long because if pullout resistance provided by roots within this range exceeds tensile strength of roots themselves, roots may fracture before shear strength of root-soil interface below can be fully mobilized. However, given the uncertainty and randomness associated with plant root growth and depth of rainfall infiltration, plants that grow quickly and have deeper root system are still preferred for soil stabilization and slope protection. Additionally, there is a considerable time effect associated with plants' solid soil slope protection. As plants age, anchoring effect of their roots on shallow slope soils inevitably exhibits dynamic changes. Therefore, understanding growth characteristics of plant roots with age under specific climatic and soil conditions is fundamental to clarifying the dynamic patterns of root-based soil stabilization effects and to scientifically and effectively utilizing plants in the prevention and control of geological disasters in shallow slope soils.

5 Conclusions

In this study, we found that in vertical pullout test, axial force of root system decreased with an increase in burial depths, but shear stress first increased and then decreased, and then both increased with increasing pullout force. Shear stress and plastic zone of surrounding soil were distributed symmetrically along root system. During non-vertical pulling, shear stress and plastic zone range of surrounding soil were no longer symmetrical. Under condition of rainfall infiltration, total displacement of root system was greater than that under natural condition, and shear stress and plastic zone of soil around roots were more likely to propagate. However, due to the difficulty in preparing samples for pullout tests, we only studied mechanism of single-root pullout resistance of taproot-type shrub plants. Future research should aim to conduct comprehensive studies of pullout mechanics of entire root system based on detailed root surveys and further numerical simulation analysis. Furthermore, in order to effectively prevent and control geological disasters in shallow soil layer of slope, it is necessary to further select a reasonable slope stability calculation method based on clarifying dynamic characteristics of plant root growth and mechanism of resistance to pullout.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

Conceptualization: LIANG Shen, LIU Yabin; Formal analysis: LIANG Shen, WANG Shu, PANG Jinghao; Funding acquisition: LIU Yabin; Investigation: LIANG Shen, WANG Shu; Methodology: LIANG Shen, LIU Yabin; Supervision: ZHU Haili, LI Guorong, HU Xiasong; Writing - original draft preparation: LIANG Shen; Writing - review and editing: LIU Yabin. All authors approved the manuscript.

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